

MODERN CRYPTOGRAPHY CHEAT SHEET

Core Concepts, Mathematics, Algorithms & Security Standards

 **Mission Control:** Confidentiality • Integrity • Authenticity | Foundation for Secure Communications

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1. CORE CONCEPTS & DEFINITIONS

TERM	DEFINITION
Plaintext (P)	Readable, unencrypted data (e.g., text, images, credit card numbers).
Ciphertext (C)	The scrambled, unreadable result of applying an encryption algorithm.
Cipher	The algorithm or mathematical method used to encrypt or decrypt data.
Key (K)	A secret bit string or value required by a cipher to transform data.
Encryption (E)	Process of converting Plaintext to Ciphertext using a Cipher and Key.
Decryption (D)	Process of reversing Ciphertext back into Plaintext using a Cipher and Key.

2. CORE SECURITY PILLARS (CIA TRIAD EQUIVALENT)

CONFIDENTIALITY

Ensures unauthorized parties cannot read or disclose transmitted data.

INTEGRITY

Ensures data has not been altered or tampered with in transit (verified via **Hash Functions**).

AUTHENTICITY

Confirms the identity of the communicating entity (verified via **Digital Certificates**).

3. HISTORICAL CIPHERS

A. CAESAR CIPHER

Mechanism: Shift Cipher where each letter is moved N places down the alphabet.

Encryption: $C = (P + Key) \bmod 26$

Decryption: $P = (C - Key) \bmod 26$

Weakness: Only 25 valid keys in English. Highly vulnerable to Brute Force and Frequency Analysis.

B. NOTABLE HISTORIC MILESTONES

Vigenère Cipher (16th C.): Polyalphabetic cipher using a repeating keyword to vary shifts.

Enigma Machine (WWII): Rotor-based electro-mechanical machine broken at Bletchley Park.

One-Time Pad / OTP (Cold War): Uses a truly random key as long as plaintext. It is the *only mathematically unbreakable cipher*.

4. ENCRYPTION CATEGORIES: SYMMETRIC VS. ASYMMETRIC

⚡ HYBRID CRYPTOGRAPHY STANDARD

- Asymmetric Encryption:** Used for Identity Verification & Secret Key Exchange.
- Symmetric Encryption:** Used for High-Speed Data Payload Encryption.

A. SYMMETRIC ENCRYPTION (PRIVATE-KEY CRYPTOGRAPHY) — SHARED SECRET KEY

CIPHER	KEY SIZE	SECURITY STATUS
DES	56-bit	Deprecated / Broken Broken in under 24 hrs in 1999.
3DES	168-bit (112-bit effective)	Deprecated (2019) Legacy support only.
AES	128, 192, 256 bits	Modern Standard Fast, secure, widely adopted.

B. ASYMMETRIC ENCRYPTION (PUBLIC-KEY CRYPTOGRAPHY) — KEY PAIR (PUBLIC & PRIVATE)

ALGORITHM	RECOMMENDED MIN KEY SIZE	KEY CHARACTERISTIC / NOTES
RSA	2048 bits (3072 / 4096)	Relies on prime factorization hardness.
Diffie-Hellman	2048 bits	Used primarily for secure key exchange.
ECC	256 bits	High Efficiency Equal security to RSA-3072 with far smaller keys.

5. MATHEMATICAL FOUNDATIONS

A. XOR OPERATION ($\oplus$ OR $\wedge$)

Returns 1 if input bits differ; returns 0 if identical.

Input A	Input B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Key Properties:

- Self-Inverse:** $A \oplus A = 0$
- Identity:** $A \oplus 0 = A$
- Commutative/Associative:** $A \oplus B = B \oplus A$

Symmetric Application:

- Encrypt:** $C = P \oplus K$
- Decrypt:** $(P \oplus K) \oplus K = P \oplus (K \oplus K) = P \oplus 0 = P$

B. MODULO OPERATION ($\%$ OR MOD)

Calculates the remainder of integer division ($X \% Y$).

Modulo Examples:

- $25 \bmod 5 = 0$ ($25 = 5 \times 5 + 0$)
- $23 \bmod 6 = 5$ ($23 = 3 \times 6 + 5$)
- $23 \bmod 7 = 2$ ($23 = 3 \times 7 + 2$)

Crucial Properties:

- Range Constraint:** For $a \bmod n$, result is always in $[0, n - 1]$.
- One-Way Function:** Given $x \bmod 5 = 4$, x has infinite solutions (4, 9, 14, 19, ...). This non-reversible property underpins asymmetric cipher security.

6. DIFFIE-HELLMAN KEY EXCHANGE (COLOR ANALOGY)

STEP	ACTION	COLOR ANALOGY	TECHNICAL MATH EQUIVALENT
1	Public Setup	Alice & Bob agree on Yellow (Public).	Agree on public parameters (g, p).
2	Secret Choice	Alice picks Red , Bob picks Blue .	Alice generates secret a , Bob generates secret b .
3	First Mix	Alice: Yellow + Red = Orange Bob: Yellow + Blue = Green	Alice: $A = g^a \bmod p$ Bob: $B = g^b \bmod p$
4	Exchange	Exchange Orange & Green publicly.	Send A & B over untrusted network.
5	Final Key	Alice: Green + Red = Brown Bob: Orange + Blue = Brown	Alice: $K = B^a \bmod p$ Bob: $K = A^b \bmod p$

✓ **Result:** Both parties derive the exact same shared secret key (K) without transmitting secrets over the network.

7. COMPLIANCE STANDARDS & SECURITY FRAMEWORKS

PCI DSS Credit card processing standard. Mandates encryption at rest & in transit.	HIPAA / HITECH US healthcare standard protecting patient medical records & privacy.	GDPR (EU) Comprehensive European law for user privacy, data protection, & rights.	DPA (UK) UK Data Protection Act governing legal processing of personal data.
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